Javvaji Venkata Koushik

javvajikoushik2004@gmail.com | +91-7816061119 | koushik-javvaji | koushikjavvaji
Portfolio | CodeForces | CodeChef

EDUCATION

Indian Institute of Information Technology, Nagpur *Nov 2022 – Jul 2026*
B.Tech in Computer Science and Engineering

- GPA: 8.18/10
- Relevant Courses: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Theory of Computation

EXPERIENCE

Pine Labs (Mosambee) – *Mumbai, India* *Sep 2025 – Sep 2026*
Software Engineer (Full-Time), Jul 2026 – Sep 2026 | Software Engineer Intern, Sep 2025 – Jul 2026

- Designed and implemented an IMPS-based Instant Payout platform with payout previews, partial/full settlements, and configurable workflows, using **Redis-backed caching** for status lookups instead of repeated DB reads; deployed to **3,000+ merchants** processing **INR 6-7 Cr daily** at **280ms p99 latency**.
- Built idempotent, retry-safe payout processing (exponential backoff, dead-letter queues) guaranteeing exactly-once settlement on API failures, with reconciliation to detect and resolve state mismatches against partner data.
- Migrated payout services from blocking RestTemplate to Spring WebFlux + Netty with async downstream calls and non-blocking processing, raising throughput from **120 to 180 requests/sec** and cutting thread usage by **50%**.
- Replaced a manual Excel-driven settlement workflow with a configurable scheduling engine (admin-defined windows, merchant/global rules, day-of-week scheduling), cutting batch processing time from **45 min to 12 min**.
- Extended Metagrox, a GraphQL reporting platform, with queries consolidating multiple REST requests into a single fetch, reducing dashboard latency from **800ms to 300ms**.
- Implemented chargeback processing and real-time transaction cancellation; improving accuracy of an AI-driven Settlement Fraud Risk Manager (SFRM) for pre-payout risk screening.

PROJECTS

DevHive — Multi-Agent AI Debugging War Room GitHub | Live

- Built a GPT-style transformer from scratch (byte-level BPE tokenizer, RoPE, grouped-query attention, SwiGLU, KV-cached inference) and trained it end-to-end; vectorized the tokenizer's pair-counting with NumPy, cutting BPE training time by **30%**.
- Designed a multi-agent debate protocol where independent LLM teammates diagnose a pasted bug concurrently, then run structured cross-critique rounds to converge on one fix; deployed live with CI on every push.
- **Technologies:** Python, PyTorch, FastAPI, WebSockets, SQLite, pytest, GitHub Actions, Render.

DRAWR — Real-Time Collaborative Drawing Platform GitHub | Live

- Built a real-time collaborative drawing platform enabling multiple users to interact simultaneously on a shared canvas with low-latency synchronization using WebSockets.
- Implemented shape tools, freehand drawing, text annotations, and undo/redo functionality with consistent state updates across concurrent clients.
- **Technologies:** Next.js, WebSockets, Express.js, Turborepo, TypeScript, Prisma, PostgreSQL, Zod.

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, JavaScript/TypeScript, SQL, Bash
- **Systems & Low-Latency:** Socket Programming, Non-blocking I/O (epoll/kqueue), Event-driven Architecture, Concurrency & Multithreading, Lock-Free Data Structures
- **Backend & Infra:** Spring Boot, Node.js, Express.js, GraphQL, Kafka, Redis, PostgreSQL, MongoDB, Docker, Kubernetes, Git, Linux

ACHIEVEMENTS AND LEADERSHIP

- **Codeforces Expert**, **4-star CodeChef**; solved **2000+** algorithmic problems across Codeforces, LeetCode, and CodeChef.
- **ICPC Amritapuri Regionalist (2025 & 2026):** Qualified for the ICPC Asia Amritapuri Regional Contest **twice**; solved **4 problems** in the 2025 regional round.
- **Meta Hacker Cup 2024:** Advanced to Round 2 and secured a **Global Rank of 3173** among **22,000+** participants worldwide.
- **TCS CodeVita 2024:** Ranked **819** in Round 1 (out of **4+ lakh** participants) and **332** in Round 2; also placed **4th** at Petrichor 2025 (IIT Palakkad).
- **Shaastra Programming Contest Finalist (2026, IIT Madras):** On-campus finalist in a national-level programming competition.
- **Problem Setter, Algorithmia '24 (IIIT Nagpur):** Authored problems for a global ICPC-style contest with **500+ teams** from **15+ countries**; mentored **100+** students as a core member of the Dot Slash Coding Club.